Contents

Chapter 1

Brownian motion in fluid flows

1.1 Introduction

Fluid flows are ubiquitous in many biological settings. The study of fluid dynamics provides insights on multiple ranges of applications. Oceanic currents and large-scale weather phenomena [13], the lift of an airplane foil [1], surfing [3] and drug biodisponibility all rely at least partly on a range of application of fluid dynamics. The latter is particularly relevant to my thesis, as it is a perfect example of the interplay between a fluid flow and Brownian motion for the diffusive transport of drugs in the bloodstream. In this Chapter, I will focus on the study of Brownian motion in fluid flows, and more specifically in viscous flows. This is a very common situation in microfluidic experiments.

1.1.1 The Reynolds number

We commonly use the Reynolds number R_e to predict the behaviour of a fluid. This number is a ratio between inertial and viscous forces in the fluid

$$R_\mathrm{e} = \frac{\rho U L}{\eta} \ , \tag{1.1}$$

where ρ is the fluid's density, U is a characteristic velocity, L is a characteristic length and η is the fluid's dynamic viscosity. Turbulence in a flow occurs at very high Reynolds number $R_\mathrm{e} \gg 1$, whereas a very small Reynolds number $R_\mathrm{e} \ll 1$ is considered laminar, with a layered and constant profile. A very classic example is that of cigarette smoke (Fig. 1.1). The smoke exists the cigarette's tip at very low velocity. By taking air density at room temperature $\rho_\mathrm{air} = 1.2\,\mathrm{kg.m^{-3}}$, viscosity at about $\eta_\mathrm{air} = 1.8\,\mathrm{Pa.s}$, and a smoke characteristic velocity $U_\mathrm{smoke} = 1\,\mathrm{cm.s^{-1}}$ and characteristic plume width $L_\mathrm{smoke} = 1\,\mathrm{mm}$,

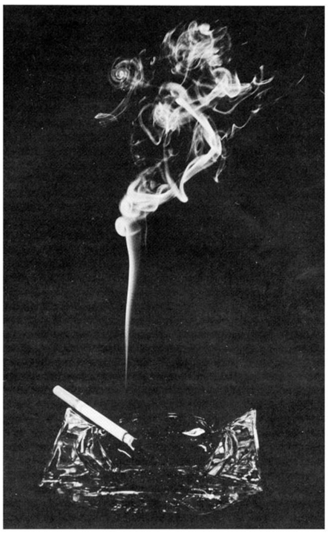

Figure 1.1: Smoke rising from a lit cigarette tip. At short distances from the tip, the smoke plume is narrow and the velocity low enough for the flow to be considered laminar. Higher up, the plume is wider and the velocity higher; the flow transitions to turbulent. Source: Physics Stack Exchange.

we can estimate the first regime at about $R_{\mathrm{e}} = 0.5$. Higher up, the smoke rises much faster and the plume is wider, with a characteristic speed of $U_{\mathrm{smoke}} = 10\,\mathrm{cm.s}^{-1}$ and a characteristic width of about $L_{\mathrm{smoke}} = 2\,\mathrm{cm}$, the ratio gives $R_{\mathrm{e}} \approx 100$. Perturbations are suppressed at low intensity due to viscous effects. As the speed rises, perturbations can no longer be damped, and the typical turbulences arise.

Most microfluidic experiments and simulations are conducted in the low-Reynolds-number regime. Despite the apparent simplicity of Stokes flows, this regime exhibits a wide range of rich phenomena, including non-trivial transport, mixing, particle migration, and microswimmer dynamics [14] [16].

In the framework of this thesis, I am interested in understanding flow effects on Brownian motion. Specifically, one question has already been very well investigated in the last century: *"What happens to a Brownian solute once it is advected by a low-Reynolds, viscous flow?"*

This Chapter first introduces the classical Taylor-Aris theory, developed to predict the dispersion of a Brownian solute under such flow. Then, I present an adjacent unpublished work carried out in collaboration with Joshua Mc. Graw, Blaise Delmotte and their respective groups, investigating hydrodynamic interactions in a dilute Brownian solute under flow advection.

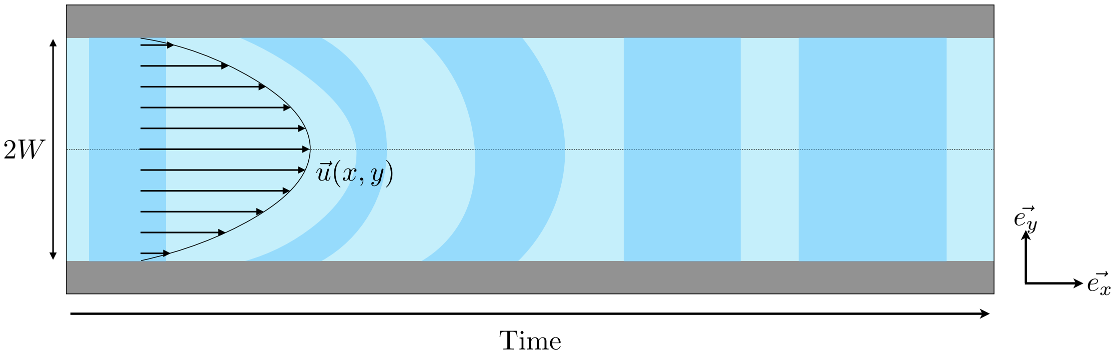

Figure 1.2: A plug of Brownian solute (darker blue) is injected in a microchannel of width $2W$. It is then advected by a viscous flow. Because of no-slip conditions at the walls (grey), the flow takes a parabolic, or Poiseuille, shape. At short times, the plug of solute takes the shape of the flow. However, because of thermal fluctuations, the plug diffuses across streamlines. This averages out the parabolic profile of the solute cross-section wise. At times larger than cross-channel diffusion time $t > \tau_\mathrm{D}$ (Eq. 1.4), the concentration profile of the solute resumes a homogeneous shape, with a larger width. This broadening of the width of the plug is quantified by the dispersion coefficient which is computed from Eq. 1.9.

1.2 Taylor-Aris dispersion

We turn our focus to the study of the evolution of a Brownian solute's concentration when advected by a viscous flow (Fig. 1.2).

We consider a microchannel of width $2W$. Fluid in that channel is advected by a unidirectional flow $u(y)\mathbf{e}_x$, and takes a characteristic Poiseuille (or parabolic shape) from the no-slip conditions at contact with the channel walls. A solute of concentration $c(x,y,t)$ and bulk diffusivity D_0 is injected in that microchannel. It obeys the advection-diffusion equation

$$\partial_t c + u(y)\partial_x c = D_0\left(\partial_x^2 c + \partial_y^2 c\right) \ . \tag{1.2}$$

We decompose

$$c(x,y,t) = \bar{c}(x,t) + c'(x,y,t), \tag{1.3}$$

where $\bar{c}$ is the cross-sectional average and c' is a small transverse correction induced by the shear. At times larger than the transverse diffusion time

$$\tau_\mathrm{D} = \frac{W^2}{D_0} \ , \tag{1.4}$$

the concentration is equilibrated across the channel width. Solving for this correction and averaging over the transverse coordinate gives an effective one-dimensional advection-diffusion equation

$$\partial_t \bar{c} + \bar{u}\, \partial_x \bar{c} = D_{\text{eff}} \partial_x^2 \bar{c} \ . \tag{1.5}$$

For a two-dimensional planar Poiseuille flow between plates separated by $2W$,

$$u(y) = \frac{3}{2}\bar{u}\left(1 - \frac{y^2}{W^2}\right) \ , \qquad -W < y < W, \tag{1.6}$$

the effective dispersion coefficient D_{eff} is then

$$D_{\text{eff}} = D_0 + \alpha \frac{\bar{u}^2 W^2}{D_0} \ . \tag{1.7}$$

The α factor arises from channel geometry and varies with its configuration. From there, we can identify a new adimensionalised quantity, the Péclet number P_{e}

$$P_{\text{e}} = \frac{\bar{u} W}{D_0} \ , \tag{1.8}$$

which compares advective to diffusive behaviour in the solute. Finally, we can write

$$D_{\text{eff}} = D_0 \left(1 + \alpha P_{\text{e}}^2\right) \ , \tag{1.9}$$

which is the classical Taylor-Aris formula [18] [19] [17] [2]. In simple words, this formula relates the dispersion of a Brownian solute in a microchannel of given width to the average fluid velocity for times beyond τ_{D}. This means that, for low enough Péclet numbers, the dispersion should be minimal, but that beyond $P_{\text{e}}^2 > 1/\alpha$, dispersion should become dominant over classical free diffusion D_0 (Fig. 1.3). It is worth noting that, while I chose to speak in terms of concentration, the same phenomenon applies for a single colloid in a viscous flow.

Taylor-Aris dispersion is still a very much ongoing field of study. Researchers have incorporated to the original Taylor-Aris theory a range of sophistications, from time-varying flow profiles [20] [12] to particle size determination [5] [6] [4] [11], and even to active swimmers [10], which will be the focus of the next Chapter.

1.3 Hydrodynamic interactions in an advected solute

The reason I became personally interested in flow effects on Brownian motion in the first place is thanks to Masoodah Gunny (ESPCI) and her thesis supervisors Alexandre Vilquin and Joshua McGraw. Their group focuses among other thematiques on the prob-

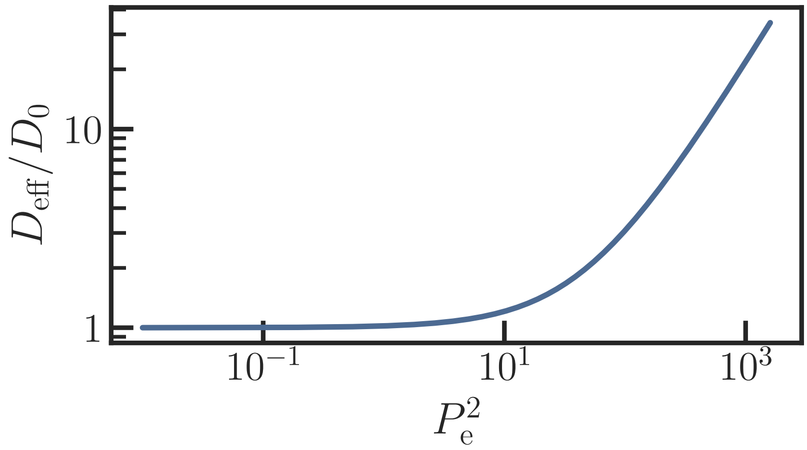

Figure 1.3: Predicted renormalised dispersion $D_{\mathrm{eff}}/D0$ against Péclet number squared P_{e}^2 computed from Eq. 1.9 with $\alpha = 1/48$, corresponding to the case of a cylindrical tube. At small P_{e}, $D_{\mathrm{eff}}/D0$ remains around 1, meaning that the dispersion effect due to advection is barely noticeable. It then crosses over around $P_{\mathrm{e}}^2 = 1$ to a second regime which is linear in P_{e}^2.

lem of Taylor-Aris dispersion near the solid-liquid interface. Specifically, they wonder in this project whether the volume fraction of a solute in a flow can influence the diffusion of this solute, even a very low density (this is akin to an ideal gas limit). Because this problem is highly nontrivial from experimental standpoint, their group together with Blaise Delmotte (LadHyX) suggested I conduct numerical simulations to replicate their system while in control of every parameter and interaction. I present here briefly their experimental protocol, their problematique and the results of our collaboration.

1.3.1 Experimental aspects

This part of the Chapter is largely based on Masoodah Gunny's PhD thesis manuscript [9].

Experiments are carried out with silica beads of radius $a = 55$nm in salted water of concentration [NaCl] $= 5.4 - 54\,\mathrm{mg.L^{-1}}$ (with corresponding Debye lengths $l_{\mathrm{D}} = 35 - 10\,\mathrm{nm}$). The solution is injected in a microfluidic channel setup of length $L = 8.8\,\mathrm{cm}$, width $W = 180\,\mu\mathrm{m}$ and thickness $H = 20\mu\mathrm{m}$. A pressure gradient ΔP is then applied between the entrance and exit of the channel to start the flowing. Very close to the wall, we can approximate the Poiseuille profile of the flow in the microchannel as a linear shear γ (Fig. 1.4), and we take the shear rate $\dot{\gamma}$ as $\dot{\gamma} = \frac{H}{2\eta}\Delta P$ with η the fluid's viscosity.

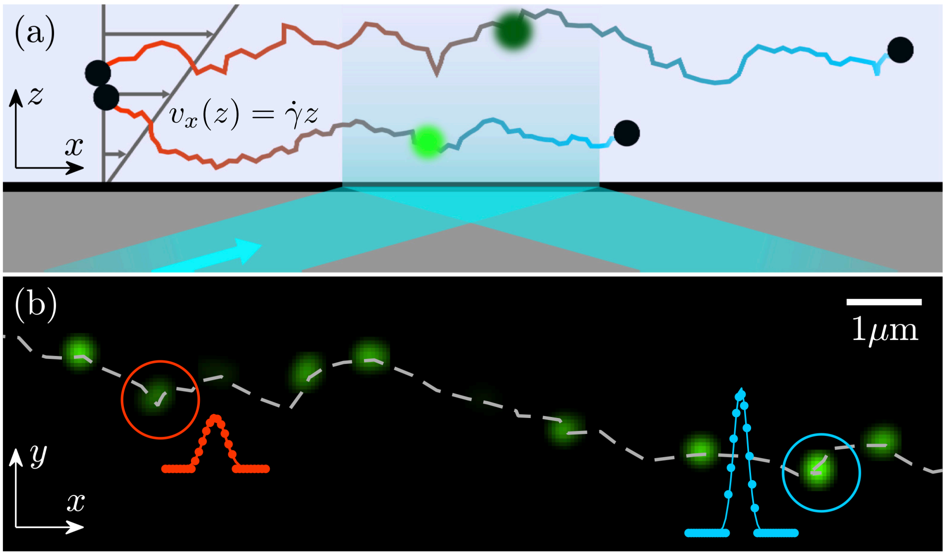

Figure 1.4: Scheme of the Taylor-dispersion TIRFM setup. (a) Two colloids of radius $a =$ 55 nm are advected by a shear flow of shear rate $\dot{\gamma}$. As they penetrate the evanescent field, they fluoresce and their positions in all three dimensions are recovered. (b) Experimental images for a single colloid. A succession of frames taken with lag time $\tau = 12.5$ ms is shown. Intensity profiles are represented in red and blue dots with Gaussian fits. The linked trajectory of the colloid is represented in a grey dashed line. Adapted from [21].

We are interested in individual colloid behaviour, and thus need an imaging technique that allows for the visualisation of individual colloids rather than a global concentration profile.

Imaging is done by Total Internal Reflection Microscopy (TIRFM) [8] (Fig. 1.5). TIRFM has the advantage of being a minimally disruptive, sub-wavelength imaging technique which still gives information on the three coordinate of a colloid's position within observation range. The principle of TIRFM relies on evanescence to collect information on the colloid's altitude. An incident laser beam hits the wall of the microchannel with incident angle θ. This angle must obey

$$\theta > \theta_c \ , \tag{1.10}$$

where θ_c is the critical angle above which no refraction is permitted, as

$$\theta_c = \sin^{-1}\left(\frac{n_1}{n_2}\right) , \tag{1.11}$$

with n_1 and n_2 the optical index of the water solution and the solid medium, respectively, and $n_1 < n_2$. Like this, refraction is prohibited, and almost all incident light is reflected. However, we know from Maxwell's equations that electromagnetic fields must remain continuous at the interface between the two media. Thus, the field must not end abruptly, but rather fade into a non-propagating electromagnetic field, which we call the evanescent wave. The penetration depth d of this wave is fairly limited, at about two colloid radii, and is given by

$$d = \frac{\lambda_0}{4\pi\sqrt{n_1^2 \sin^2\theta - n_2^2}} , \tag{1.12}$$

where λ_0 is the vacuum wavelength of the laser. This field then excites fluorescent molecules at the surface of the colloids, and we retrieve the emitted light. We can infer the distance between a colloid and the interface z from the intensity a colloid emits, as

$$I(z) = I_0 \exp\left(\frac{-z}{d}\right) , \tag{1.13}$$

where I_0 is the maximal intensity at the interface.

Snapshots of the solution under advection are taken with a camera. Trajectories are then linked from individual successive frames. From the trajectories, displacement distributions are computed, and the transverse diffusion coefficient is then defined as

$$D_y = \frac{\sigma_y^2}{2\tau} , \tag{1.14}$$

where σ_y^2 is the variance of the distribution of displacements in the y-direction (along the channel's wall, perpendicular to the flow), and τ is the time step between two consecutive frames, and is taken as $\tau = 1\,\text{ms}$.

Then, solutions of various concentrations of silica colloids in salted water are tested. It is more intelligible to speak in terms of volume fraction ϕ, that is to say, total volume occupied by colloids, rather than concentration from there on. We define the volume fraction as

$$\phi = \frac{N4/3\pi a^3}{WLH} , \tag{1.15}$$

where N is the total number of colloids in solution. The transverse diffusion coefficients D_y computed from Eq. 1.14 are then recovered for various values of $\phi \in [5 \times 10^{-6}; 4 \times$

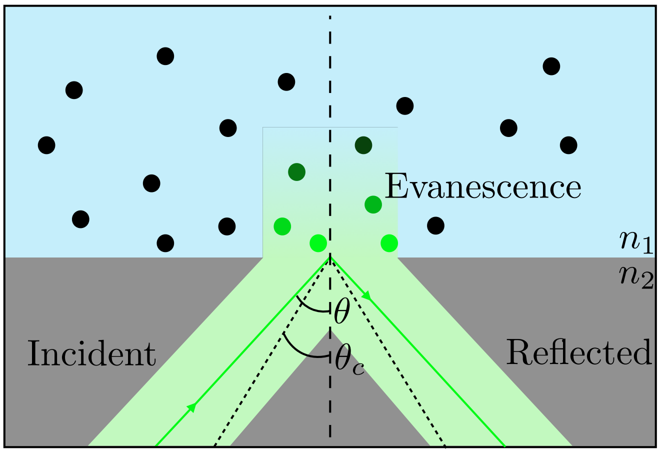

Figure 1.5: TIRFM experimental principle. An incident beam hits the interface with an angle $\theta > \theta_c$. It is almost entirely reflected away from the interface. A small portion of the beam, however, does penetrate the sample in the form of an evanescent wave. The colloids in the sample can be visualised owing to their fluorescence when they are illuminated by the evanescent wave and their altitude is then recovered from the light intensity they emit.

10^{-5}] and $\dot{\gamma}$ and shown in Fig. 1.6.

Because we work with extremely dilute solutions (below the limit of $\phi = 0.05$ above which the viscosity of the fluid becomes dependant on particle concentration [7]), and as hydrodynamic interactions decay in $\propto 1/r$ inter-colloid, and $\propto 1/r^3$ between a colloid and a wall, we should not expect any visible change on the dynamics of the system within that range of volume fraction. However, experimental results seem to point towards a shear rate and volume fraction dependence on D_y (Fig. 1.6, bottom panels). Indeed, for high enough volume fractions ($\phi = 3.7 \times 10^{-5}$, or 0.000037% of total volume occupied by colloids), D_y becomes larger than the bulk diffusion coefficient D_0 at $D_0 = 3.9\,\mu\text{m}^2.\text{s}^{-1}$, and this effect is exacerbated by higher shear rates.

As puzzling as this result is, there are a few potential causes that can easily be ruled out by carrying out numerical simulations. This has already been done for the case of electrostatic interactions by Élodie Millan as one of her thesis projects.

Numerical methods

As our collaborators in ESPCI hypothesised a hydrodynamic effect causing the enhancement in transverse diffusion D_y, I conduct all the forthcoming simulations using the RigidBlobWalls [15] again. This package, as a reminder, is optimised for the simulation

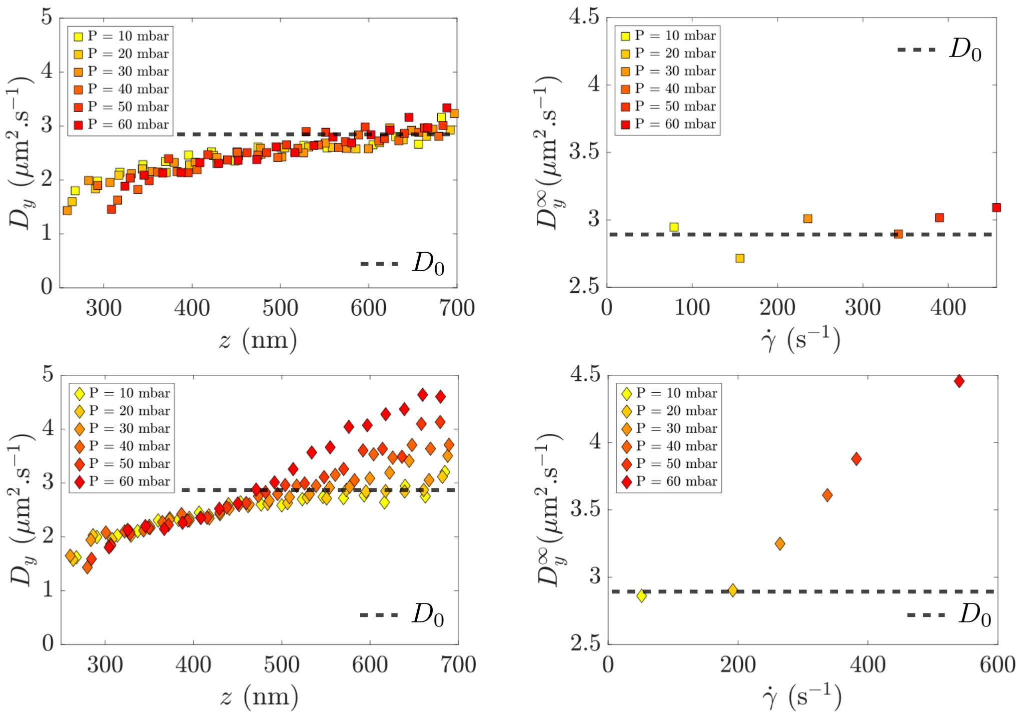

Figure 1.6: Transverse diffusion coefficients computed for $\phi = 5.4 \times 10^{-6}$ (top) and $\phi = 3.7 \times 10^{-5}$ (bottom) and for various pressure gradients ΔP or equivalent shear rates $\dot{\gamma}$ (in colours). Left panels show the diffusion at binned heights z above the channel wall. We notice a strong dependence of D_y on z in both cases, with a drop in diffusion as the colloid diffuses closer to the wall, which is to be expected from Brenner and Faxen's laws. However, once the colloid moves away from the wall, D_y picks up and may even reach higher values than the bulk value $D_0 = 3.9\mu\text{m}^2.\text{s}^{-1}$ (in horizontal dashed lines). This effect seems even more noticeable with higher $\dot{\gamma}$, as shown on the bottom right panel, where D_y^∞ indicates the diffusion coefficient very far from wall interactions. Adapted from [9].

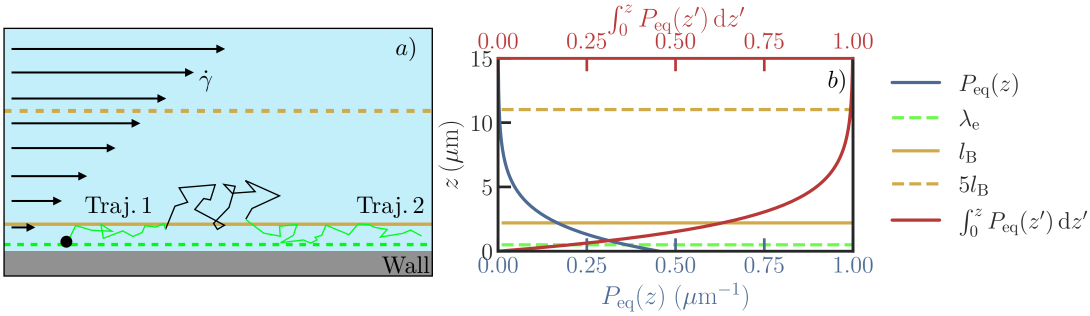

Figure 1.7: Simplified numerical simulation setup. **a)** Colloids are initialised in position in a box of dimensions $L_x \times L_y \times 5l_{\mathrm{B}}$, with l_{B} the Boltzmann constant. The box is periodic in the x and y directions, and has a confining plane at $z = 0$. A shear flow of shear rate $\dot{\gamma}$ is applied in the x direction. Trajectories are recorded for a total time of $T = n_{\mathrm{steps}} \times \Delta t$. Trajectories crossing the evanescent limit λ_{e} are killed, and trajectories re-entering below λ_{e} are tracked as new trajectories if they exist for a long enough time. Data treatment is then done to extract diffusion coefficients from these trajectories. **b)** Equilibrium distribution of a perfect gas of colloids in solution, as predicted by the Gibbs-Boltzmann distribution (blue, Eq. 1.16) and corresponding cumulative sum (red). The evanescent limit λ_{e} is represented in green. This limit is the maximum height at which experimentalists can retrieve data from the colloids. The Boltzmann length l_{B} is represented in a full yellow line. We estimate that 99% of the colloids lie under $5l_{\mathrm{B}}$ (yellow dash line), which is why we choose this height as the upper bound to ensure minimal evaporation of the colloids, and thus a well-defined volume fraction.

of colloids in hydrodynamic and electrostatic interactions, and allows both wall effects and shear flows. The task involves reproducing the TIRFM setup numerically, generating trajectories in the very same conditions as the experiments, and extract similar quantities from these trajectories. The point is to obtain data from a perfectly controlled environment, with perfect tracking of every colloid's position, and with all interactions known.

Simply put, a simulation has four main steps:

Parametrisation. The user provides the physical parameters (buoyant mass of a colloid Δm, colloid radius a, shear rate $\dot{\gamma}$, fluid viscosity η, thermal energy $k_{\mathrm{B}}\Theta$, Debye length l_{D}, amplitude of the electrostatic repulsion potential A, evanescent limit λ_{e}) and simulation parameters (time step Δt, total number of steps n_{steps}, periodic lengths L_x and L_y). This is done using the `inputfile.dat` and `parameters.py` scripts. The `parameters.py` script then reads `inputfile.dat` to convert, calculate, and pass parameters to the other scripts.

As we discussed in the previous paragraphs, the two key quantities we must have full control over are the shear rate $\dot{\gamma}$ and the volume fraction ϕ. The former is simply given as an input to the code in `inputfile.dat`, whereas the latter must be input indirectly in the form of total number N of *blobs*, which in this case are equivalent to colloids.

Initialisation. `sample_initial_positions_gb.py` reads from `parameters.py` the array of ϕ values to be probed. The volume fraction is actually ill-defined in our system if considered without care. Indeed, the simulation code is built to either work in triply periodic boundaries, or doubly periodic in (x, y) and half-infinite in z with a confining plane at $z = 0$. This means the maximum height above which colloids do not escape is set physically, not through user input directly. We can only decide the initial positions of the colloids, but without proper care once the system reaches equilibrium, colloids diffuse in the vertical direction to preferential positions. We recall from Chapter 3 that the distribution of the positions in a perfect gas solution is predicted from the Gibbs-Boltzmann equation

$$P(z) = \frac{1}{Z} \exp\left(\frac{U(z)}{k_\mathrm{B}\Theta}\right) \ , \tag{1.16}$$

with Z a normalisation factor, and $U(z)$ the total potential energy, as

$$U(z) = A \exp(-\frac{z}{l_\mathrm{D}}) + \frac{z}{l_\mathrm{B}} \ , \tag{1.17}$$

with A the dimensionless magnitude of the electrostatic repulsion potential, l_D the Debye length and l_B the Boltzmann constant. All these quantities are defined in Chapter 3. The distribution of colloids can be integrated to show that about 99% of the colloids lie under $5l_\mathrm{B}$, and about 60% below l_B (Fig. 1.7b)). Thus, the problem is twofold: if we choose to lower the simulation cost by using a small box of height H_box with $H_\mathrm{box} < 5l_\mathrm{B}$ to place our colloids at the beginning of the simulations, the solution will inevitably homogenize to a lower volume fraction after some simulation time (Fig. 1.8). Furthermore, one cannot simply choose a volume fraction for the whole box and homogeneously place the colloids in that box. Colloids diffuse in the vertical direction and reach the Gibbs-Boltzmann distribution, which is not homogeneous (Fig. 1.7 b)). Thus, the volume fraction is higher close to the wall and lower far from it (Fig. 1.7 b)). To compute the true volume fraction ϕ_{5l_B} we must feed the code for the whole $5l_\mathrm{B}$ slab in order to retrieve the target ϕ within the λ_e slab, we use

$$\phi_{5l_\mathrm{B}} = \frac{\phi\lambda_\mathrm{e}}{5l_\mathrm{B}\frac{1}{Z}\int_0^{\lambda_\mathrm{e}} P_\mathrm{eq}(z)\mathrm{d}z} \ . \tag{1.18}$$

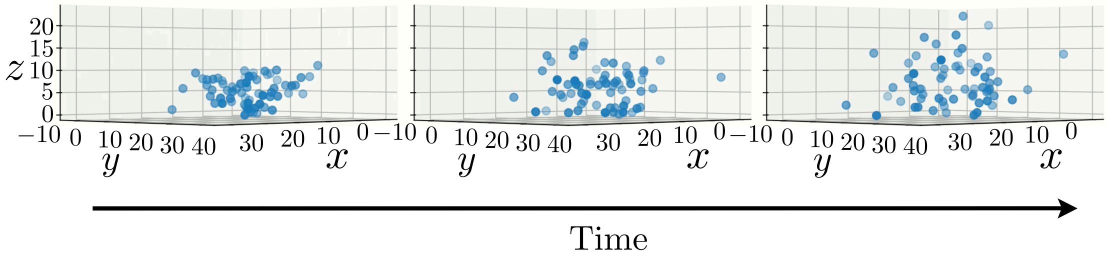

Figure 1.8: Numerical simulation results at initialisation and through time, for a bad initialisation of the system. Colloids were laced in a 10 µm tall box and left to thermalise with $l_{\mathrm{B}} = 1.1 \times 10^{-2}$ m. As expected, colloids escape the boundaries of the original box from thermal fluctuations, thus effectively reducing the volume fraction inside the box.

Then, we may simulate the whole $5l_{\mathrm{B}}$ slab and fill it with the adequate amount of colloids to reach the desired ϕ. Given that, for our system, $5l_{\mathrm{B}} = 5.5 \times 10^{-2}$ m, and the TIRFM observation window is of 22 µm × 22 µm, we estimate the total amount of blobs in a simulation at $N = 3.7 \times 10^{7}$ to reach a reasonable target volume fraction $\phi = 4 \times 10^{-4}$. This is clearly much too high, and we must find a workaround.

A very simple way of doing so is to artificially lower l_{B}. By taking a much higher value for the gravitational acceleration, at $g = 5000 \times g_{\mathrm{Earth}}$ where $g_{\mathrm{Earth}} = 9.81\,\mathrm{m.s^{-2}}$, the new Boltzmann constant is $l_{\mathrm{B}} = 2.20 \times 10^{-6}$ m, which in turn gives a simulation of $N = 152$ colloids. This is much more reasonable.

`sample_initial_positions_gb.py` creates, in the current working directory, a new directory for each value of ϕ. In each new directory, it writes all the simulation scripts. Finally, and most importantly, it computes the initial colloid positions within the simulation box from the Gibbs-Boltzmann distribution of Eq. 1.16 using a simple exclusion algorithm, and writes them to `sphere_array.clones`. This step is crucial to ensure that the system is prethermalised, and avoids a potentially costly prethermalisation step. This script can also be used to verify that the Gibbs-Boltzmann distribution is respected and that the system is properly thermalised.

Simulation. The script `multi_bodies.py` calls the entire simulation pipeline and launches the shear flow on the pre-thermalised colloidal suspension. It writes the trajectories at each time step to `run_blobs.sphere_array.config`. Other relevant simulation data, such as failed iterations and total computation time, are written to auxiliary output files.

Data treatment. From the trajectories stored in `run_blobs.sphere_array.config`, the trajectories are parsed, and the particle displacements are computed.

We take care to ensure we limit our studies to the same field of view as the experiments use. This means that colloids may escape the evanescent wave limit λ_e beyond which experimentalists cannot retrieve data do exist in simulation, but that we kill in the treatment process their trajectories once they escape λ_e. Furthermore, if they re-enter the field of view below λ_e for long enough time durations, their trajectories are treated as new trajectories by the treatment code. MSDs are computed from displacements on each trajectory and then binned by average height.

Diffusion quantities are then extracted from the MSD if, and only if, a diffusive regime is identified, namely beyond the transverse diffusion time τ_D and when the log-log fits yield a time exponent equal to 1.

All parametrisation, initialisation, and treatment codes are available on appendix. Simulation codes are available on the GitHub repository of the authors `https://github.com/stochastichydrotools/rigidmultiblobswall`.

Results

Finally, we can compare the results of the simulations to the experimental results. We compute the renormalised transverse diffusion coefficient D_y/D_0 as a function of average trajectory height for various values of ϕ and $\dot{\gamma}$ (Fig. 1.9). We notice that, in contrast to the experimental results, no clear dependence of D_y on ϕ or $\dot{\gamma}$ is visible. The overall tendency of D_y/D_0 shows a slight increase close to the contact with the wall at $z + a =$ 55nm, and quickly reaches a saturation around 1. This is in agreement with the expected behaviour of a Brownian solute in a viscous flow, and with Brenner and Faxen's laws. The absence of any clear dependence on ϕ and $\dot{\gamma}$ suggests that the experimental results may be due to an artefact, and that the observed enhancement of D_y may not be due to hydrodynamic interactions. This is in line with the fact that, at such low volume fractions, hydrodynamic interactions should be negligible. However, it is worth noting that experiments and numerical data were treated differently. Indeed, in experiments, trajectories were first linked from different frames, and displacements were computed from these trajectories. In simulations, we directly access the whole labeled trajectories, thus ruling out any potential tracking artefact.

Furthermore, the simulations were run with a much higher value of g than in experiments, which may have an influence on the results. However, we do not expect this to be the case, as the Boltzmann distribution of colloids in the vertical direction is well respected in simulations, and thus we are confident that the system is properly thermalised and interactions should not be strongly modified.

Finally, it is worth noting that the diffusion coefficient D_y itself has different definitions between our simulation results and the experiments. In experiments, D_y is computed from the variance of the distribution of displacements in the y-direction at a fixed time step τ, as shown in Eq. 1.14. This means that there is no proper verification of a diffusive regime before extracting a diffusion coefficient. If the system is not in a diffusive regime, the extracted coefficient may not be meaningful. In simulations, we compute D_y from the slope of the MSD in the diffusive regime, which is a more standard definition of diffusion coefficient.

Ongoing work is still being conducted to understand the origin of this discrepancy. We are currently analysing numerically generated TIRFM-like videos, which can be found on my GitHub repository ADD THE GIT, with the same tracking and data analysis as the experiments used. If the effect then becomes visible, it would suggest that the effect is due to a data treatment artefact. However, if the effect is not visible, more investigations would be needed. We could for instance verify the actual volume fractions in experiments to fully rule out hydrodynamic effects.

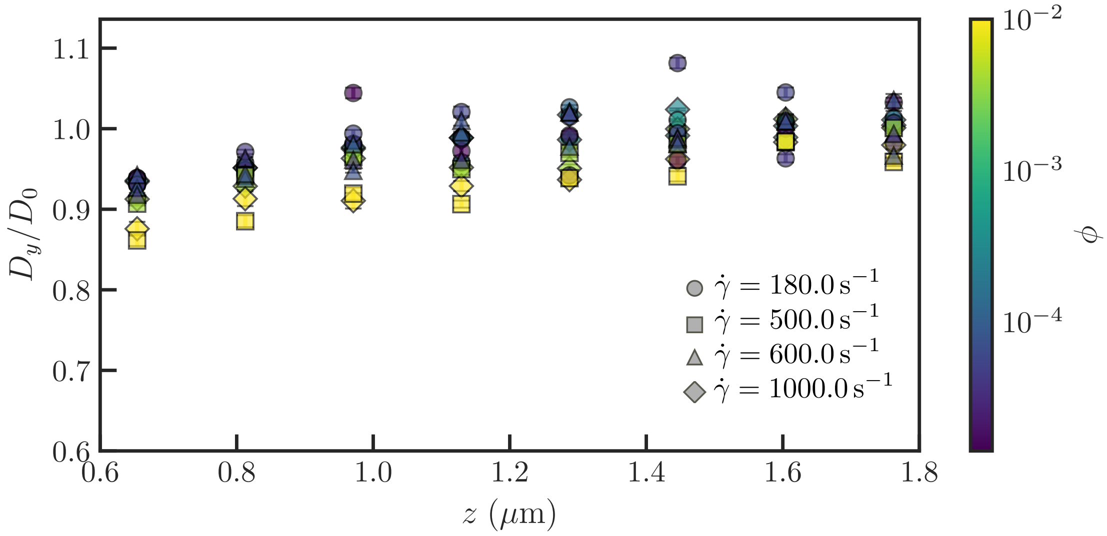

Figure 1.9: Renormalised transverse diffusion coefficient D_y/D_0 as a function of average trajectory height. Simulations were run once for each $(\phi, \dot{\gamma})$ couple and for $N_{\mathrm{steps}} = 50000$, with time step $\Delta t = 0.0001\,\mathrm{s}$, fluid viscosity $\eta = 0.001\,\mathrm{Pa.s}$, colloid radius $a = 55\,\mathrm{nm}$, buoyant mass $\Delta m = 1.92 \times 10^{-16}\,\mathrm{kg}$, gravity acceleration $g = 5000 \times 9.81\,\mathrm{N.m}^{-2}$, thermal energy $k_{\mathrm{B}}\Theta = 4.14 \times 10^{-21}\,\mathrm{J}$, and cutoff evanescent limit $\lambda_{\mathrm{e}} = 1800\,\mathrm{nm}$. The colormap represents volume fractions. Disks correspond to shear rate $\dot{\gamma} = 100.0\,\mathrm{s}^{-1}$, squares to $\dot{\gamma} = 500.0\,\mathrm{s}^{-1}$, triangles to $\dot{\gamma} = 600.0\,\mathrm{s}^{-1}$ and diamonds to $\dot{\gamma} = 1000.0\,\mathrm{s}^{-1}$. The overall tendency of D_y/D_0 shows a slight increase close to the contact with the wall at $z + a = 55\mathrm{nm}$, and quickly reaches a saturation around 1. No clear effects of the shear rate are visible.

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